# My robotics workbook

Build something, test a robot and explain what you noticed. Four teams of three share two robots. Keep your own ideas and let each teammate try every job.

### Robot jobs

Driver: edit and start after the lane is clear. Navigator: predict, place the stopped robot and reset its payload. Observer: watch, measure and record one number, tick or phrase. Keep hands away while the robot moves.

### Build jobs

Builder connects pieces. Designer finds parts and checks the plan. Tester tries the model and suggests one improvement. Everyone builds. Switch jobs when your mentor calls time.

### Our agreement

We ask before Run, respect the lane, share the parts and controls, and stop when something is wrong. A surprising result gives us a new question. Our models stay in labeled team trays.

Team ID _____ Student IDs A _____ B _____ C _____

# Welcome to robotics

Today you will make a robot move and build something for it.

### Meet your team

Meet your mentors and teammates. Everyone will get time to build and try the robot. Your mentor will help you with each new step.

### Watch a robot in action

Watch the demonstration. You can share something you notice, ask a question or simply watch.

### Try an idea together

Choose a signal for your robot and make a removable mascot. Try a small change and see what happens.

### Help each other

Take turns with the controls and parts. Check that the lane is clear before starting. Anyone can call stop. Keep hands away while the robot moves.

There is time to explore, ask for help and try again.

# Week 1 Meet your robot

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A robot welcome signal

**Predict together** Number these actions in run order: drive, show Y, wait. What changes if the first and last lines trade places?

Our prediction _____

```
await signal("Y")
await pause(500)
await drive(90)
```

Read this fragment. Use the full project supplied by your mentor.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A removable robot mascot

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about instruction order? What did you notice when the robot ran?

Next idea _____ Saved as _____

# Week 2 Delivery to a target

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A 40 cm delivery

**Predict together** Measure C = ____ cm around our wheel. Predict 180 degrees: ____ cm; 360 degrees: ____ cm. Explain using half and one full turn.

40 cm target: (40 / C) × 360 = _____ wheel degrees. Round at the end.

```
wheel_degrees = 180
await drive(wheel_degrees)
await signal("D")
```

Predict cm = (degrees / 360) × C. Use our measured C and the full supplied project.

| Run | Wheel degrees | Predicted cm | Measured cm |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A cargo tray that keeps its load

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about reaching a target? What did you observe in your distance tests?

Next idea _____ Saved as _____

# Week 3 Robot dance studio

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A repeating robot dance

**Predict together** How many drives? How many turns? Does the final D repeat? Which indentation tells you?

Our prediction _____

```
for beat in range(3):
    await drive(90)
    await turn(90)
await signal("D")
```

Read this fragment. Use the full project supplied by your mentor.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A dancing character with moving parts

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about repeating actions? What did you notice during the robot dance?

Next idea _____ Saved as _____

# Week 4 Parking assistant

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** Parking beside a wall

**Predict together** At a 150 mm threshold, classify 220, 150, 90, and -1. Which readings mean near? Which means check the sensor?

Our prediction _____

```
stop_mm = 150
gap = distance_sensor.distance(port.C)
if 0 <= gap <= stop_mm:
    await signal("P")
```

Read this fragment. Use the full project supplied by your mentor.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A garage with a dependable target wall

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about sensors and decisions? What did you observe as the robot neared the wall?

Next idea _____ Saved as _____

# Week 5 The floor tells the robot

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A floor marker delivery stop

**Predict together** If dark reads 12 and light reads 76, is 44 a useful threshold? What happens if the threshold is below both readings?

Our prediction _____

```
threshold = 40
reading = color_sensor.reflection(port.D)
if reading < threshold:
    await signal("M")
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A road with a delivery marker

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn about detecting a marker? What did you observe on light and dark surfaces?

Next idea _____ Saved as _____

# Week 6 A rescue attachment

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A supply box rescue

**Predict together** Why might two short side guides keep a cube in front better than one flat beam? How could too much contact slow the robot?

Our prediction _____

```
async def deliver():
    await signal("R")
    result = await seek_marker(threshold)
    print(result)

# inside student_recipe
await deliver()
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A rescue scoop or guide

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn while building your guide? What did you observe when it moved the cube?

Next idea _____ Saved as _____

# Week 7 Community robot mission

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A community helper prototype

**Predict together** Which part of your mission is most likely to fail? What small test could answer that before you run everything?

Our prediction _____

```
mission = "marker"
if mission == "marker":
    result = await seek_marker(threshold)
else:
    result = await park(stop_mm)
print(result)
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A small community mission scene

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What did you learn while combining your mission? What did you observe in your first complete test?

Next idea _____ Saved as _____

# Week 8 Robot showcase

Team ID ____ Student IDs A ____ B ____ C ____ Date _____

**Robot challenge** A short public demonstration

**Predict together** Before the demo, predict the stop reason. What will you explain if the robot stops for a limit instead of detecting its target?

Our prediction _____

```
await signal("Y")
result = await seek_marker(threshold)
print("Our result:", result)
```

Read this fragment. Use the full project supplied by your mentor.

If using the marker: dark ____ light ____ threshold ____. Parking: start gap ____ mm.

| Run | One change | Prediction | Observed result |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

**Build** A working exhibit for the showcase

Hand-test result _____ One change we made _____

Driver turns: A [ ] B [ ] C [ ] Each child handled and observed: [ ]

Share aloud: What can you explain or do now? What did you notice in today's demonstrations?

Next idea _____ Saved as _____

# What I can explain now

Student ID _____ Date _____

Answer independently before the Week 8 briefing. Your mentor may read the questions without giving answers.

**1.** A recipe says: show Y, drive, show D. Which action happens second?

_____

_____

**2.** wheel_degrees = 180 is changed to wheel_degrees = 360. If everything else stays the same, do you predict more travel, less travel, or exactly the same travel? Why?

_____

_____

**3.** A loop repeats drive then turn three times. How many drive actions happen?

_____

_____

**4.** The robot parks when a valid gap is 150 mm or less. Should it park at 100 mm? Should it park at 200 mm?

_____

_____

I can explain a robot program: 1 Not yet / 2 A little / 3 With help / 4 Mostly / 5 Confidently

More robotics? Yes / Maybe / No / Prefer not to answer. My next project idea: _____

# Reading the robot recipe

Edit the small student_recipe section in the complete project. Helpers, ports and safety limits are prepared by mentors. Ask before changing them.

| Example | Meaning |
|---|---|
| await signal("Y") | Show a letter |
| await pause(500) | Wait half a second |
| await drive(180) | Turn the wheels half a rotation |
| wheel_degrees = 180 | Give a number a name |
| for beat in range(3): | Repeat the indented actions three times |
| if reading < threshold: | Choose an action when the comparison is true |
| await deliver() | Run the named action |

### Small repairs

Keep quotation marks around letters. Keep the colon after a loop or decision. Preserve indentation. Wheel degrees are not centimeters or the robot’s heading. Measure the real result.